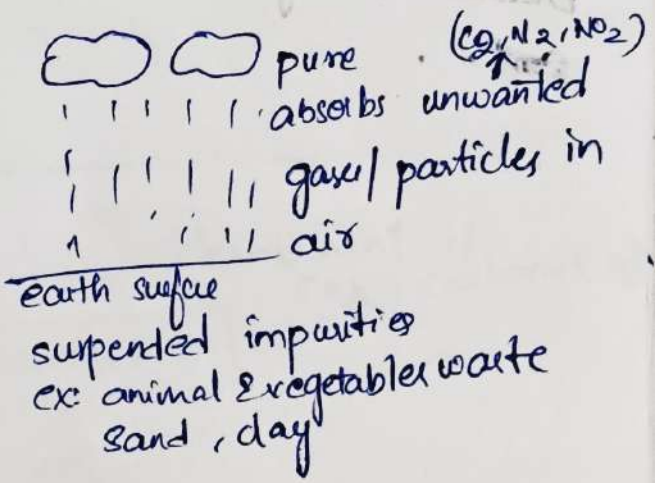
air → 12–15% Water
Human → 60% "
Vegetables 90–95% "

Basic: food
water
air
shelter

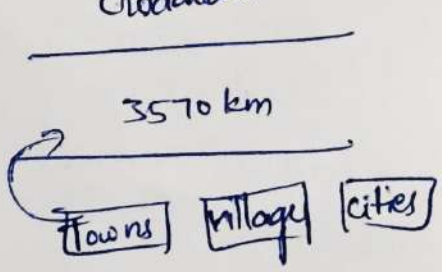
Sources of water

- Surface water sources (above ground level)
  - Rain W
  - River W
  - lake W
  - Sea W
- Sub surface (or) underground WS (below ground)
  - Dwells
  - Springs

① Rain Water:-
purest form of water is rain w. due to heat of sun during evaporation.



② River water:
primary source of water
highly polluted
highly turbid → colourless, odourless, tasteless
self purification → removal of impurities by its own



contamination with garbage
~~bacteria~~

③ Lake / Pond water

stagnant at 1 place only

low dissolved oxy & high salts

contaminate with bacteria

lenting → flowing

lotting → stagnant

High salts → $CaCl_2$, $MgCl_2$

④ Sea / Ocean water:

largest source of water

unutilized

↳ due to presence of salts

brakish → salty taste

Total 3.5 % ⇒ 2.6 % NaCl

$CaCl_2$, MgO, $Ca(HCO_3)_2$, KCl } 0.9 %

contaminate with salts

Soft water:
The water which produces lather with the soap is called soft water.

Hard water
Water which does not give lather or produce a very little amount of lather is called Hard water. (due to presence of salts)

Hardness:-
lather preventing property of water is called hardness.

Hardness
- Temporary H (Carbonate H)
- Permanent H (Non carbonate H)

$\boxed{\text{Total H} = \text{Temporary H} + \text{Permanent H}}$

Hardness causing salts:
bicarbonates of (Ca, Mg) → $Ca(HCO_3)_2$ (162) mol. wt
chlorides → $CaCl_2$ (111), $MgCl_2$ (95), $Mg(HCO_3)_2$ (146)
sulphates → $CaSO_4$ (136), $MgSO_4$ (120)
} responsible for Hardness

Temporary H. Causing salts	Permanent H. Causing salts
bicarbonates of Ca, Mg	chlorides & sulphates of Ca, Mg
$Ca(HCO_3)_2$	$CaSO_4$ $CaCl_2$
$Mg(HCO_3)_2$	$MgSO_4$ $MgCl_2$

Removing H.

Tempolary H : $Ca(HCO_3)_2$, $Mg(HCO_3)_2$ $\xrightarrow{\text{boiling}}$ $CaCO_3$ (Insoluble)

Permanent H $CaCl_2$ $CaSO_4$ $MgCl_2$ $MgSO_4$ $\xrightarrow{\text{zeolite process}}$ ion exchange method, lime soda process.

Tempolary H:

$$Ca(HCO_3)_2 + \text{heat} \longrightarrow \underset{\text{insoluble}}{CaCO_3} + H_2O + CO_2$$

$HCO_3 \rightarrow CO_2$

Permanent H:-

Soap is sodium / potassium salts of higher fatty acids like stearic, oleic and palmitic acids.

$$\underset{\text{Na Soap, Na stearate}}{2C_2H_5COONa} + \underset{\text{P.H}}{CaCl_2} \longrightarrow \underset{\text{Ca stearate, insoluble}}{(C_2H_5COOCa)} + 2NaCl$$

$$2C_2H_5COONa + MgSO_4 \longrightarrow (C_2H_5COOMg) + 2Na_2SO_4$$

Units of Hardness:-

1. ppm → parts per million
2. mg/L → milli gram / litre
3. °Cl → degree clark
4. °Fr → degree french

1) ppm
The number of parts by weight of $CaCO_3$ equivalents of H. causing salt (P) in 1 million parts of water
(10^6/10 lakh)

2) mg/L
The amt of the $CaCO_3$ equivalents hardness causing salts in milligrams (P) in 1 litre of water

(3) °Cl
No of parts of $CaCO_3$ equivalents of H causing salts in 70,000 parts of water

(4) °Fr.
No. of parts of $CaCO_3$ equivalent H in 10^5 parts of water

Relation :-

$$\boxed{1\ ppm = 1\ mg/L = 0.07^\circ Cl = 0.1^\circ Fr}$$

$$\boxed{1^\circ Cl = 1.43^\circ Fr = 14.3\ ppm = 14.3\ mg/L}$$

Q) A sample of hot water contains 120 mg/L of hardness. Express the hardness of water in ppm °Cl, °Fr

$120\ mg/L = 120\ ppm = 8.4^\circ Cl = 12^\circ Fr$

°Cl 0.07×120
0.7×12
8.4

°Fr 120×0.1
$= 12$

Q) A water sample found to contain Hardness of 80° Cl. Express the Hardness of water in mg/L, °Fr and ppm.

$80°Cl = 114.4°Fr = 1144\ ppm = 1144\ mg/L$

1.43×80
14.3
114.4

14.3×80
1143
1144

Degree of Hardness $CaCO_3 \rightarrow$ insoluble

Hardness of the H. causing salts in terms of $CaCO_3$ equivalence

$$= \frac{\text{amt of H. causing salt} \times 100}{\text{Molwt of H.C.S}}$$

mol wt of $CaCO_3$

H.C.S → Hardness Causing salt

Q) A sample of hardwater contains the following dissolved salts per litre. ~~$CaCO_3 = 111$~~ $CaCl_2 = 111\ mg/L$, $CaSO_4 = 1.36\ mg/L$, $Ca(HCO_3)_2 = 16.2\ mg/L$, $Mg(HCO_3)_2 = 14.6\ mg/L$, Silica = 40 mg/L and turbidity = 10 mg/L. Calculate temporary, permanent and total hardness of the water sample.

S.No	HCS	Amt of HCS (mg/L)	mol wt of HCS	Hardness in terms of $CaCO_3$ $= \frac{A \times 100}{\text{mol wt}}$
1	$CaCl_2$ P.H	111	111	$\frac{111 \times 100}{111} = 100$
2	$CaSO_4$ P.H	1.36	136	$\frac{1.36 \times 100}{136} = 1$
3	$Ca(HCO_3)_2$ T.H	16.2	162	$\frac{16.2 \times 100}{162} = 10$
4	$Mg(HCO_3)_2$ T.H	14.6	146	$\frac{14.6 \times 100}{146} = 10$
5	Silica (not H.C.S)	40		
6	Turbidity	10		

Temporary Hardness

$Ca(HCO_3)_2 + Mg(HCO_3)_2$

$= 10 + 10$

$= 20$ mg/L

permanent Hardness

$CaCl_2 + CaSO_4$

$= 100 + 1$

$= 101$ mg/L

Total Hardness = Temporary H + permanent H

$= 20 + 101$

$= 121$ mg/L

Q) Calculate Temporary, permanent & total H of water sample. The analysis of water is as follows. $Ca(HCO_3)_2 = 16.2$ ppm, $Mg(HCO_3)_2 = 14.6$ ppm, $CaSO_4 = 13.6$ ppm, $MgSO_4 = 12$ ppm, $MgCl_2 = 9.5$ ppm.

S.No	HCS	Amt of HCS (mg/L)	Mol wt of HCS	Hardness of $CaCO_3$ eq $= \frac{A \times 100}{mol.wt}$
1	T.H $Ca(HCO_3)_2$	16.2	162	$H = \frac{16.2 \times 100}{162} = 10$
2	T.H $Mg(HCO_3)_2$	14.6	146	$H = \frac{14.6 \times 100}{146} = 10$
3	P.H $CaSO_4$	13.6	136	$H = \frac{13.6 \times 100}{136} = 10$
4	P.H $MgSO_4$	12	120	$H = \frac{12 \times 100}{120} = 10$
5	P.H $MgCl_2$	9.5	95	$H = \frac{9.5 \times 100}{95} = 10$

Temporary Hardness

$Ca(HCO_3)_2 + Mg(HCO_3)_2$

$= 10 + 10$

$= 20$ mg/L

permanent Hardness

$CaSO_4 + MgSO_4 + MgCl_2$

$= 10 + 10 + 10$

$= 30$ mg/L

Total Hardness = T.H + P.H

$= 20 + 30$

$= 50$ mg/L

Q) Calculate T.H, P.H & Total H of water containing per litre. $MgCl_2$ = 19 mg/L, $MgSO_4$ = 60 mg/L, NaCl = 36.5 mg/L, $CaCl_2$ = 11.1 mg/L, $Ca(HCO_3)_2$ = 32.4 mg/L, $Mg(HCO_3)_2$ = 7.3 mg/L

S.No	HCS	~~Amt of~~ HCS (mg/L)	mol wt of HCS	$H = \frac{A \times 100}{mol\ wt}$
1	$MgCl_2$	19	95	$\frac{19 \times 100}{95} = 20$
2	$MgSO_4$	60	120	$\frac{60 \times 100}{120} = 50$
3 Not a salt	NaCl	36.5	~~58.5~~	
4	$CaCl_2$	11.1	111	$\frac{11.1 \times 100}{111} = 10$
5	$Ca(HCO_3)_2$	32.4	162	$\frac{32.4 \times 100}{162} = 20$
6	$Mg(HCO_3)_2$	7.3	146	$\frac{7.3 \times 100}{146} = 5$

Temporary H

$Ca(HCO_3)_2 + Mg(HCO_3)_2$

$= 20 + 5$

$= 25$

permanent H

$MgCl_2 + MgSO_4 + CaCl_2$

$= 20 + 50 + 10$

$= 80$

Total H = Tem H + Per H

$= 25 + 80$

$= 105$ mg/L

Drinking water – W.H.O parameters

→ should be colourless, odourless, tasteless

→ turbidity less than 10 ppm

→ not contain poisonous metals (Pb, As, Cr)

→ pH range 7-8.5

→ TH → less than 500 ppm

→ free from disease causing micro-organisms

Impurities of water

- Suspends impurity
  - Inorganic: Sand, clay
  - organic: Animal & vegetable waste
- Biological I
- Dissolved impurities
  - Colloidal I

* River water

Dissolved I :- completely dissolve in the form bicarbonates, chlorides & sulphates of Ca & Mg → Hardness

Sodium & potassium salts → alkalinity

Gases responsible for (CO_2, NO_2, SO_2) etc for odour & acidity

if O_2 is (P) → corrosion

Colloidal I

finely divided impurities

Silica, Aluminum hydroxide, bacteria

Biological I

disease causing - plants, algae, fungi & pathogens.

matter

Softening Techniques

The removal of hardness causing salts (HCS) from the hard water is called softening.

~~technique~~

Simply softening conversion of hard water to soft water

not for drinking purpose contain salts

1. lime soda process
2. Zeolite process
3. Ion exchange method

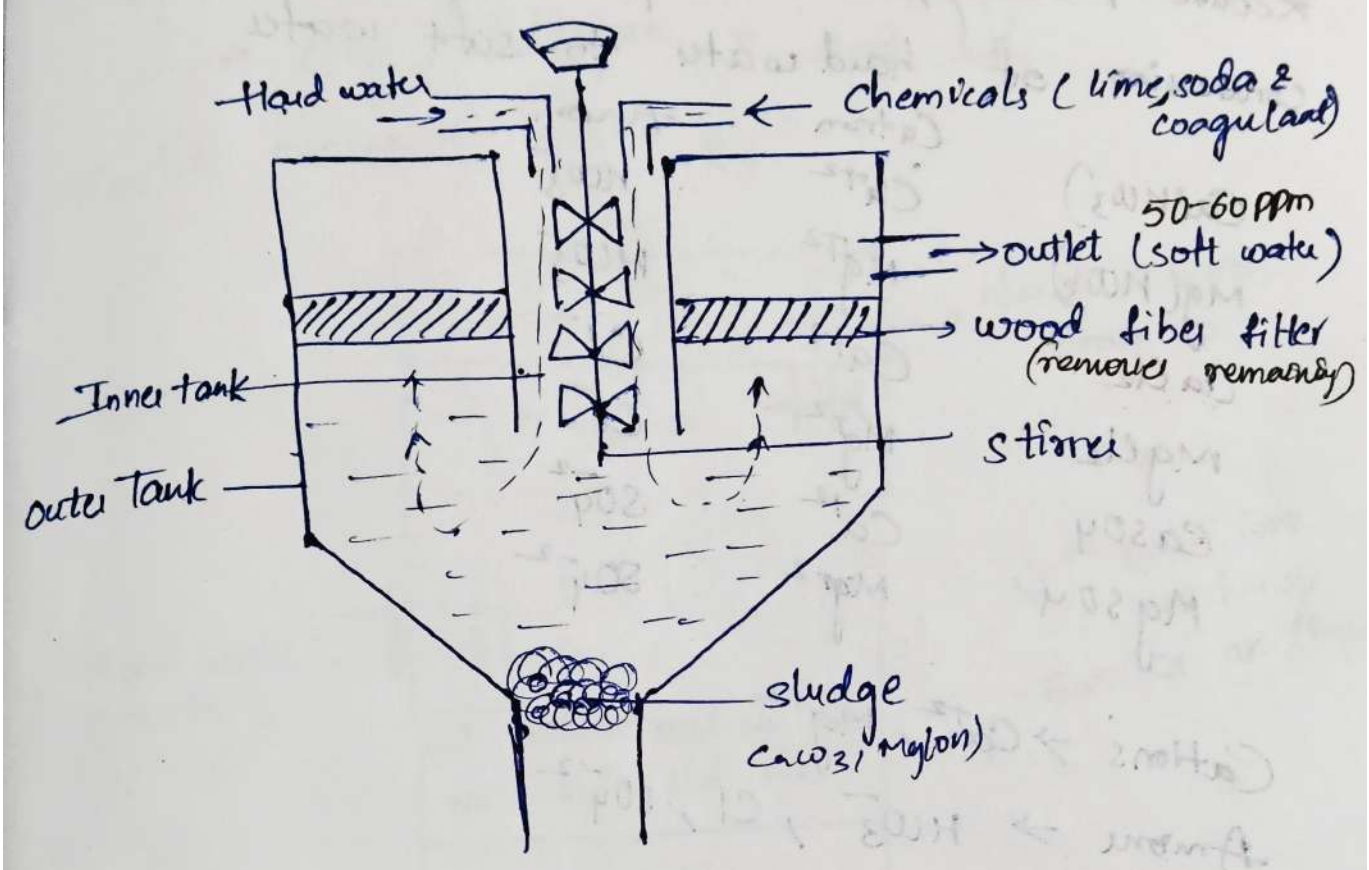
1. Lime Soda process:

1. Cold
2. Hot

1. Cold LSP:	2. Hot LSP:
principle: Calculated amt of lime & soda + hard water + coagulent (in rate of xn) ($NaAlO_2$) (S. Aluminate) $\xrightarrow{RT}$ soft water + sludge (waste) (insoluble ppt) $Mg(OH)_2$, $CaCO_3$	principal → Calculated amt of lime & soda + hard water $\xrightarrow{110-150°C}$ soft water + sludge (insoluble ppt) $CaCO_3$, $Mg(OH)_2$

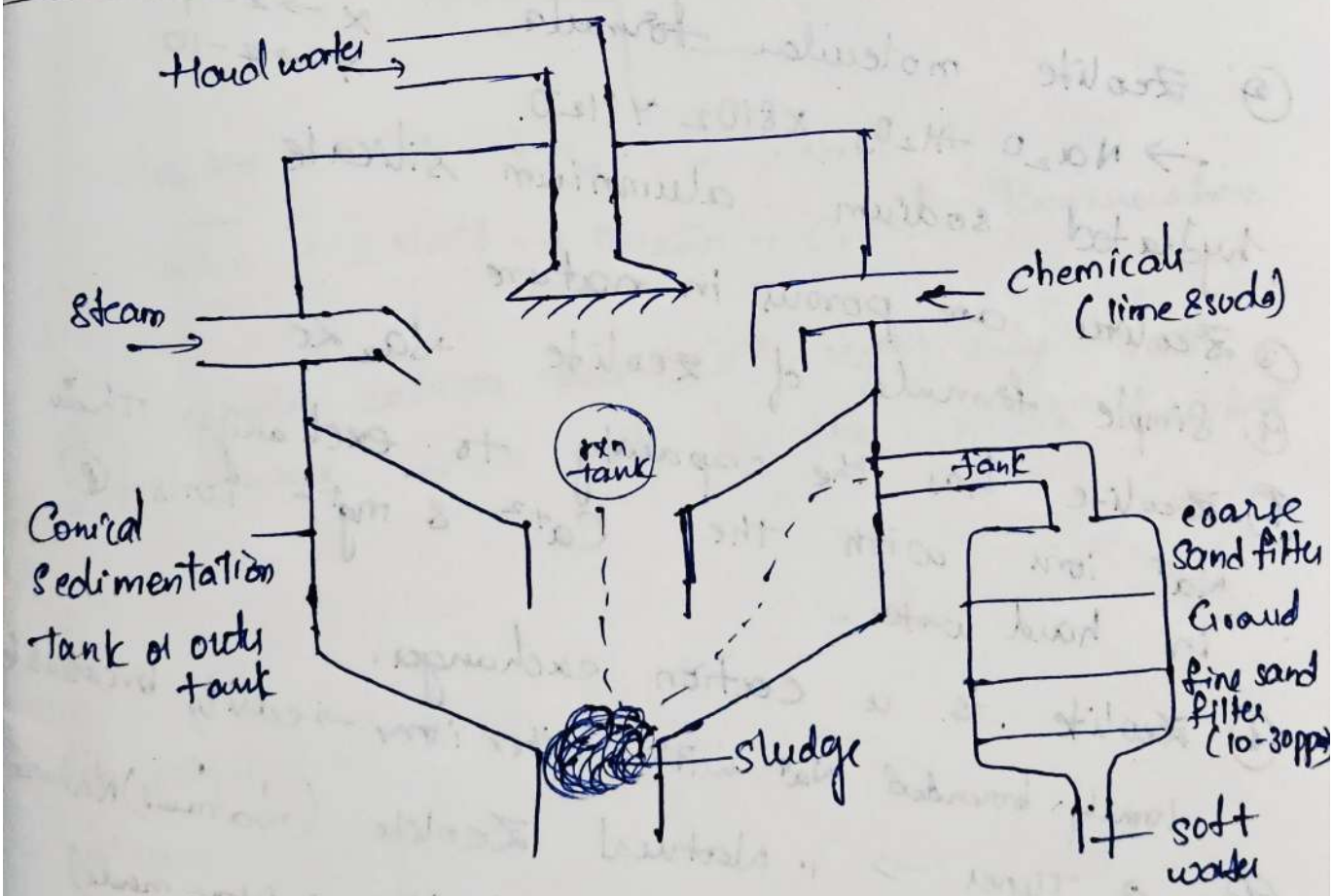
lime → $Ca(OH)_2$

soda → Na_2CO_3



drinking H2O → 0 – 2 ppm

cold LSP → 50 - 60 ppm



Zeolite Method/Permutit process:-

conversion of hard water to soft water

	Cation	Anion
$Ca(HCO_3)$	Ca^{+2}	HCO_3^-
$Mg(HCO_3)$	Mg^{+2}	HCO_3^-
$CaCl_2$	Ca^{+2}	Cl^-
$MgCl_2$	Mg^{+2}	Cl^-
$CaSO_4$	Ca^{+2}	SO_4^{-2}
$MgSO_4$	Mg^{+2}	SO_4^{-2}

Cations → Ca^{+2}, Mg^{+2}

Anions → HCO_3^-, Cl^-, SO_4^{-2}

① Zeolite = Zeo + lite
boiling stones

② Zeolite molecular formula $x \to 2-4$, $y \to 2-10$
→ $Na_2O\ Al_2O_3\ xSiO_2\ yH_2O$
hydrated sodium aluminium silicate

③ Zeolites are porous in nature

④ Simple formula of zeolite Na_2Ze

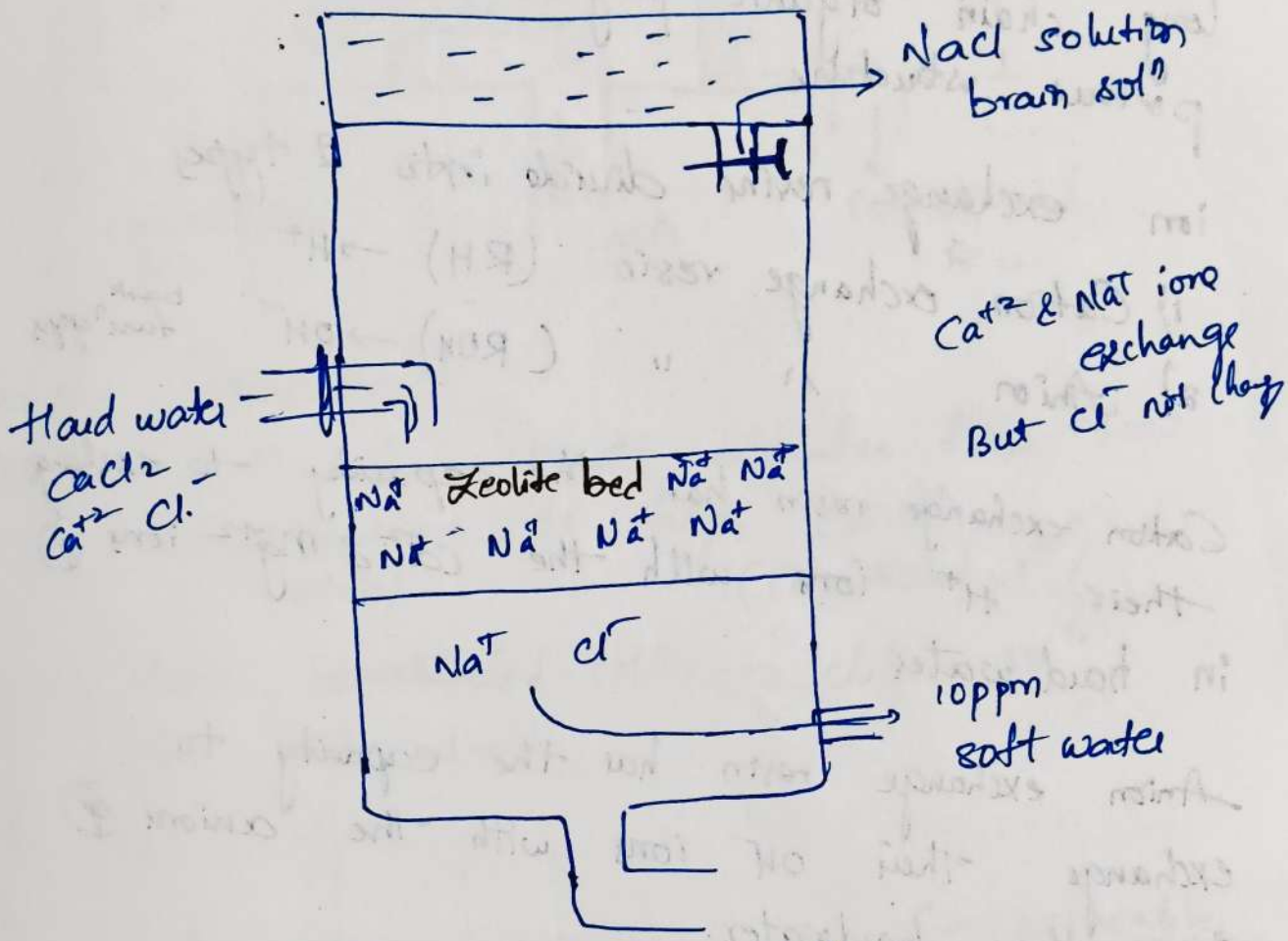
⑤ Zeolite has the capacity to exchange their Na^+ ions with the Ca^{+2} & Mg^{+2} ions in hard water.

⑥ Zeolite is a cation exchanger.
loosely bounded Na^+ with other ions → easily breakable

⑦ 2 Types → 1. Natural Zeolite (Natural) Nature
2. Synthetic Zeolite (Man made)

⑧ Synthetic Zeolites ~~are~~ ~~non~~ prepared from china clay, feldspar & soda ash.

Zeolite process



$Na_2Ze + CaCl_2 \rightarrow CaZe + 2NaCl$

$CaZe + 2NaCl \rightarrow Na_2Ze + CaCl_2$ Regeneration.

only remove cations ions (Ca^{+2}, Mg^{+2}) only but not anions → so it is the main drawback

→ Non porous

→ porous

Ion exchange method / demineralization method / deionization process:

Ion exchange resins are insoluble cross linked long chain organic polymers with a micro porous structure

ion exchange resins divide into 2 types

1) Cation exchange resin (RH) → H^+
2) Anion " " (ROH) → OH^- basic func^n grps

Cation exchange resin has the capacity to exchange their H^+ ions with the Ca^{+2} & Mg^{+2} ions in hard water

Anion exchange resin has the capacity to exchange their OH ions with the anions in the hardwater.

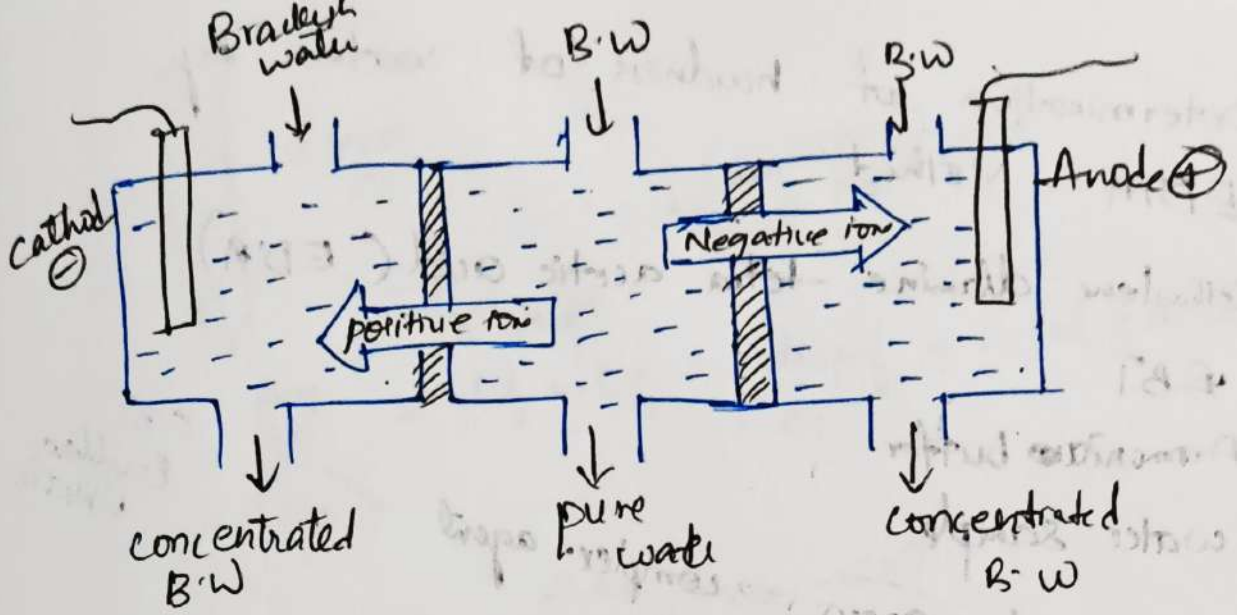
1) COOH, SO_3H
2) NH_2, NH

Desalination of Brackish (ocean) (sea) H_2O

The process of removal of salts from the blackish water is klas desalination

2 Methods

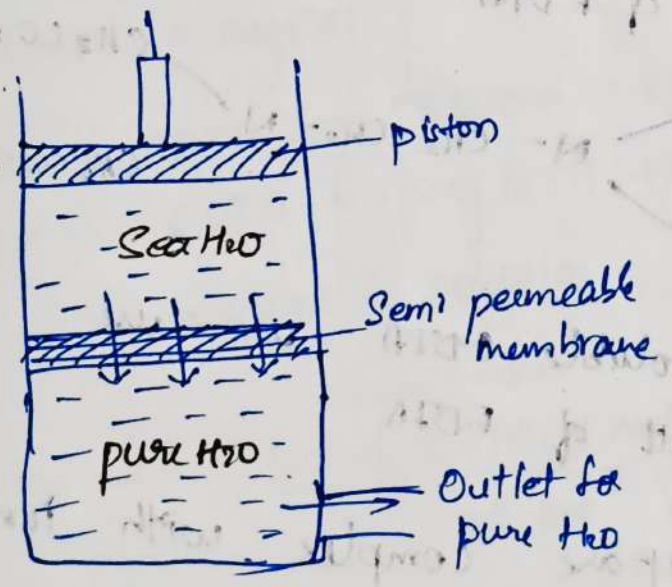
1. Electro dialysis
2. Reverse osmosis



Reverse Osmosis :-

Osmosis - flow of solvent molecules from low concentration (dilute) to high conc^n (concentrated sol^n).

R.O : from concentrated sol^n to dilute sol^n (or) high → low



Semi permeable membrane
↓
allow solvent molecules

Sea water (impure) $\xrightarrow{\text{solvent}}$ pure H_2O (pure)

Determination of hardness of water by EDTA Method :-

Ethylene diamine tetra acetic acid (EDTA)

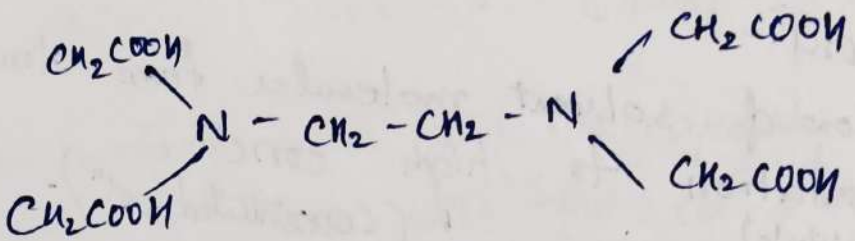
EBT

Ammonia buffer

water sample

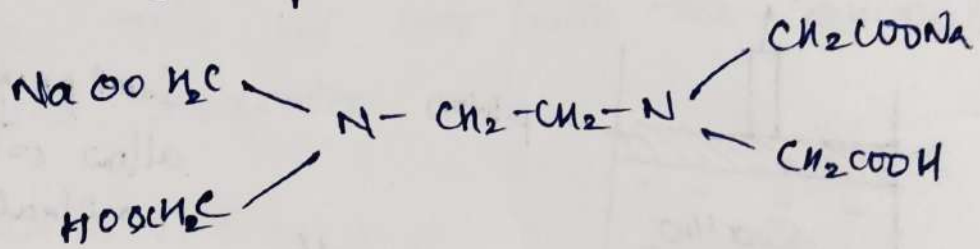
EBT
Erythro
Bretin

structure of EDTA → complex agent



EDTA
insoluble in water

disodium salt of EDTA



→ due to insoluble EDTA we use disodium salt of EDTA

→ EDTA → prepare complex with ion (P) in H_2O.

→ EBT → sky blue → it gives colour to sol^n → indicator

* Ammonia buffer
[NH4Cl + NH4OH] → to maintain the pH of sample H_2O
always the water sample pH - 8 to 10
(Mg^{+2}, Ca^{+2})

* water sample + EDTA → complex product

EDTA soln

Ca^{+2} & Mg^{+2} → hardness causing metal ions

20 ml of W.S + 2-3 drops of EBT + 2-3 ml of A. Buffer

$$Ca^{+2}/Mg^{+2} + EBT \xrightarrow[pH=10]{A.Bu} Ca\,EBT \text{ / } Mg\,EBT$$

wine red complex (unstable)

$$\xrightarrow{+\,EDTA} Ca\,EDTA + EBT$$
$$Mg\,EDTA + EBT$$

colourless, Blue colour

$Ca^{+2} + EBT \rightarrow Ca\,EBT$

$Mg^{+2} + EBT \rightarrow Mg\,EBT$

$Ca\,EBT + EDTA \rightarrow Ca\,EDTA + EBT$

$Mg\,EBT + EDTA \rightarrow Mg\,EDT + EBT$

wine red colourless colourless Blue colour.

$$\text{Total hardness} = \frac{\text{Vol of EDTA (Given in table)} \times \text{molarity of EDTA (0.01M)} \times 100 \times 1000}{\text{vol of water sample (20/10 ml)}}$$

mg/L 0-75 → soft H_2O

75 - 150 → moderate hard H_2O

150 - 300 → hard H_2O

above 300 very hard water

} Given by WHO

Determination of Alkalinity experiment present in the water sample.

Alkalinity causing ions → OH^-, CO_3^{-2}, HCO_3^-
carbonate bicarbonate

pink → colourless } phenopthalene Alkalinity / partial A
only 2 OH^-, CO_3^{-2}

+ 2-3 drop of M.O
yellow ~~pink~~ → pink ~~yellow~~ } methyl orange Alkalinity / Total A
measure all 3 OH^-, HCO_3^-, CO_3^{-2}

$H^+ + OH^- \rightarrow H_2O$
$H^+ + CO_3^{-2} \rightarrow HCO_3^-$ } ph A
$H^+ + HCO_3^- \rightarrow H_2O + CO_2$ } M.O A

Boiler Troubles:- made of (iron)

boiler → closed vessel → used in industries, power generation, cooking

produce steam

1. generation of electricity
2. sterilization
3. mechanical effort
4. energy storage
5. treatment of concrete

if hot water taken in boiler

Major boiler troubles

Boiler Troubles
- Scale & Sludge
- Caustic embrittlement
- priming & foaming
- Boiler corrosion

Water for steam generation

should not have suspended / oil impurities

dissolved org or inorg compo.

dissolved oxy & CO_2

colour, odour; pH nearer to 7

hardness more than 0.2ppm

~~not~~ caustic alkalinity in range of 0.15 to 0.45.

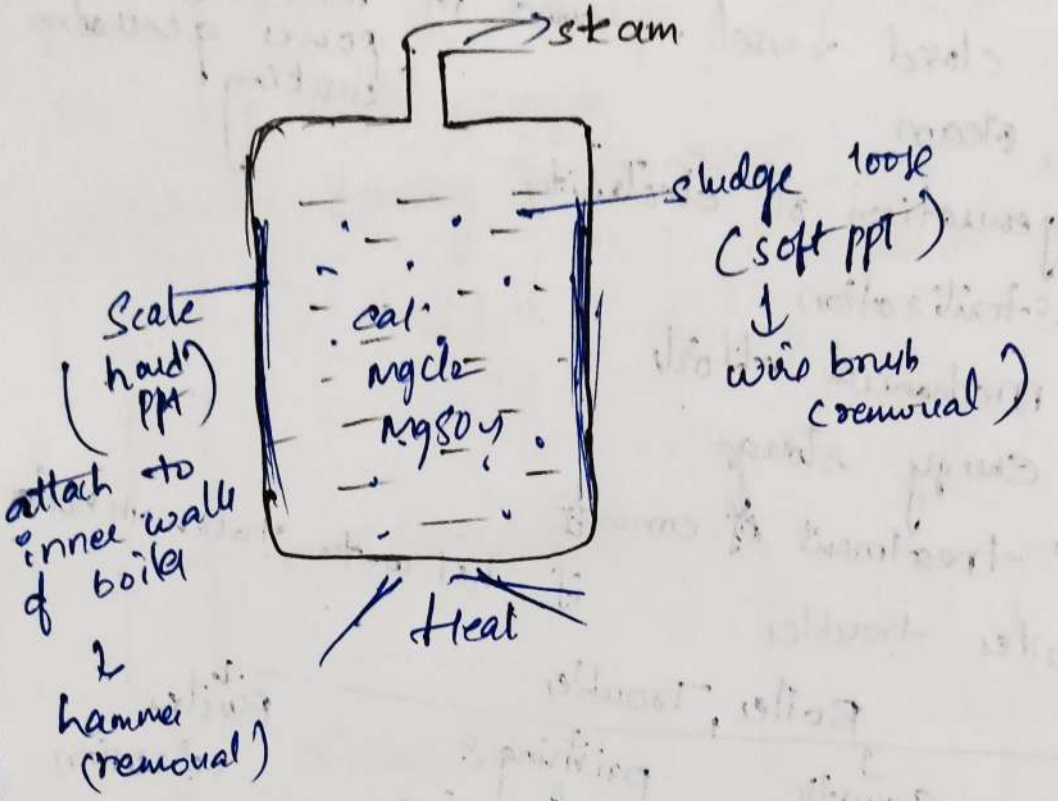
① Scale & sludge:-

Sludge:- Sludge is a soft, loose and slimmy ppt formed within the boiler

ex:- $MgCO_3$, $MgCl_2$, $CaCl_2$, $MgSO_4$ etc

Scale:- Scale is a hard substance which is very strongly attached to the inner walls of the boiler.

ex: $CaSO_4$, $CaCO_3$, $Mg(OH)_2$



$Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2$
scale

$CaSO_4 \rightarrow$ scale

$MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 + 2HC$
scale

$Mg(OH)_2$ scale

$CaSiO_3$ & $MgSiO_3 \rightarrow$ Scale

disadvantage of scale

1) fuel wastage
2) reduce the efficiency
3) may explode - cracks
4) cause choking of pipes
5) degradation of boiler materials
increase of risk of accident

Removal

1) using scrapper - wirebrush
2) thermal shock
3) using chemicals - 5-10% HCl & by adding EDTA

prevention of scale

1. Internal conditioning / Internal Treatment
2. External conditioning / external "

1. phosphate C : adding Na phosphat
Carbonate C : " Na carbonate
Calogen C : " calogen
↓ sodium hexa meta phosphate

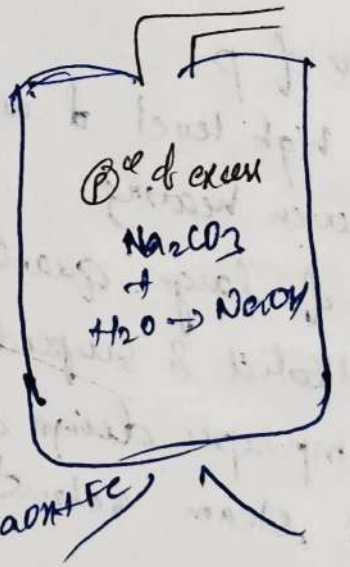
$Na_2[Na_4(PO_3)_6]$

cleaning of washing machine drums

2) Caustic embrittlement :- (NaOH, cracks)

$Fe + NaOH \rightarrow Na_2FeO_2 + H_2$

Sodium ferrate



Prevention of c.e :-

i) use phosphate salts instead of Na Carbonate

ii) use Na_2SO_4 / agar agar gel ~~components~~ compounds to fill the fine cracks

2) priming & foaming

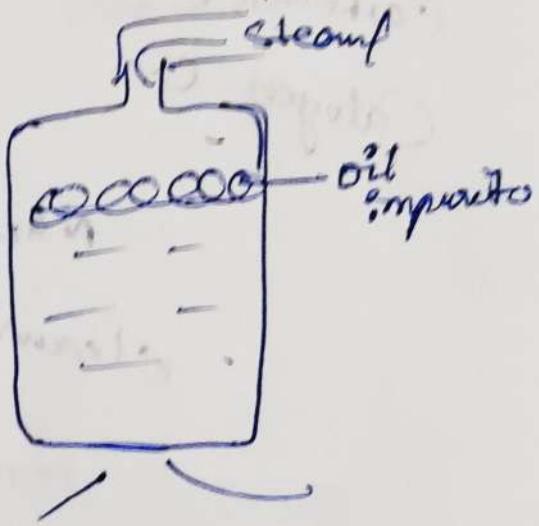
priming :- the process of formation of lead steam is called priming

foaming :- the process of formation of continuous foam / bubbles in boilers.

priming :-
if suspend impurities are ①

along with steam some impurities are also came ::

→ priming → carryover bubble prevent bed steam



→ foaming → along with steam foam is formed

Normal bubble

Causes of p

very high level of water

uneven heating

Ⓒ of large quantity of dissolved salts, org matter, alkalies & suspended matter &

improper design of boiler

↑ steam velocity.

Prevention

maintaining proper water level

even temp

proper design

4) Boiler Corrosion:-

degradation / destruction of boiler materials (A) due to chemical or electrochemical dissolved gases & salts

due to dissolved O_2, CO_2, salts